

Proposition (statement)

A Proposition or Statement is a declarative sentence that is either true or false, but not both.

For eg. 1)

Three plus three equals six T
and Three plus three equals seven F

So both are statements.

2) " $x+y > 1$ " is not a statement.

The truth or falsity of a statement is called its truth value.

NOTE: Questions, exclamations and commands are not propositions.

For eg:

- a) The sun rises in the west. ✓
- b) $2+4=6$ ✓
- c) $(5,6) \subset (7,6,5)$ ✓
- d) Do you speak Hindi? X
- e) $4-x=8$ X
- f) Close the door. X
- g) What a hot day! X

To represent simple propositions by letters $p, q, r, \dots$ known as proposition variables.

Propositional variables can only assume two values: 'true' denoted by T or 1 and 'false' denoted by F or 0.

Connectives: The words and phrases (or symbols) used to form compound propositions are called connectives.

There are five basic connectives called Negation, Conjunction, Disjunction, Conditional and Biconditional.

The following symbols are used to represent connectives.

Nature of the compound statement	Connective word	Symbol used	Symbolic form
Negation	not	$\sim, \neg$	$\sim p$
Conjunction	and	$\wedge$	$p \wedge q$
Disjunction	or	$\vee$	$p \vee q$
Conditional	if ---- then	$\rightarrow$	$p \rightarrow q$
Bi-conditional	if and only if	$\leftrightarrow$	$p \leftrightarrow q$

Truth Values of Compound Propositions using Different Connectives.

Conjunction: If p and q are two statements, then conjunction of p and q is the compound statement denoted by $p \wedge q$.

$p \wedge q$ is true when both p and q are true otherwise it is false.

for ex.

a) Kolkata is in Bengal and Delhi is in India. (T)

b) Lucknow is capital of U.P. and $3+2=6$ (F)

c) London is in China and $4+3=7$ (F)

d) $7+4=8$ and $5+10=7$. (F)

Table 1: Truth Table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction: If p and q are two statements, the disjunction of p and q is the compound statement denoted by $p \vee q$.

The statement $p \vee q$ is true if at least one of p or q is true. It is false when both p and q are false.

For eg:

- a) Kolkata is in Bengal or Delhi is in India. (T)
- b) Lucknow is capital of U.P. or $3+2=6$ (T)
- c) London is in China or $4+3=7$ (T)
- d) $7+4=8$ or $5+10=7$. (F)

Table 2: Truth Table for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Negation: If p is any proposition, the negation of p , denoted by $\sim p$. This proposition is false when p is true and true when p is false.

for eg:

1) If p : Delhi is capital of India.
 $\sim p$: Delhi is not capital of India.

2) q : $3+2=7$
 $\sim q$: $3+2 \neq 7$

Table 3: Truth value of $\sim p$

p	$\sim p$
T	F
F	T

Ex Construct a truth table for each compound proposition.

i) $p \wedge (\sim q \vee q)$

(ii) $\sim(p \vee q) \vee (\sim p \wedge \sim q)$

Sol (i)

p	q	$\sim q$	$\sim q \vee q$	$p \wedge (\sim q \vee q)$
T	T	F	T	T
T	F	T	T	T
F	T	F	T	F
F	F	T	T	F

(ii)

p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim(p \vee q)$	$(\sim p \wedge \sim q)$	$\sim(p \vee q) \vee (\sim p \wedge \sim q)$
T	T	F	F	T	F	F	F
T	F	F	T	T	F	F	F
F	T	T	F	T	F	F	F
F	F	T	T	F	T	T	T

Logical Equivalence:

Two propositions $P(p, q, \dots)$ and $Q(p, q, \dots)$ are said to be logically equivalent or simply equivalent or equal if they have identical truth tables and is written as

$$P(p, q, \dots) \equiv Q(p, q, \dots)$$

As shown below truth tables of $\sim(p \wedge q)$ and $\sim p \vee \sim q$ are identical. Hence $\sim(p \wedge q) \equiv \sim p \vee \sim q$

p	q	$p \wedge q$	$\sim(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

$\sim(p \wedge q)$

p	q	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

$\sim p \vee \sim q$

Algebra of Propositions:

Propositions satisfy various laws which are listed below:

Idempotent laws

1a) $p \vee p \equiv p$

1b) $p \wedge p \equiv p$

Associative laws

2a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$

2b) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

Commutative laws

3a) $p \vee q \equiv q \vee p$

3b) $p \wedge q \equiv q \wedge p$

Distributive Laws

4a) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

4b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

Identity Laws

$$5a) p \vee F \equiv p$$

$$5b) p \wedge T \equiv p$$

$$6a) p \vee T \equiv T$$

$$6b) p \wedge F \equiv F$$

Complement Laws

$$7a) p \vee \sim p \equiv T$$

$$7b) p \wedge \sim p \equiv F$$

$$8a) \sim T \equiv F$$

$$8b) \sim F \equiv T$$

Involution Law

$$9) \sim(\sim p) \equiv p$$

De Morgan's Laws

$$10a) \sim(p \vee q) \equiv \sim p \wedge \sim q$$

$$10b) \sim(p \wedge q) \equiv \sim p \vee \sim q$$

Ex. Use truth table to prove the distributive law

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Sol.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Since the entries in the 5th and last column of the table are the same, the two propositions are logically equivalent.

Conditional Proposition:

If p and q are proposition, the compound proposition "if p then q " denoted by $p \rightarrow q$ is called a conditional proposition. Here the proposition p is called hypothesis, and the proposition q is called the conclusion.

The only way by which $p \rightarrow q$ is false when p is true and q is false, otherwise its always True.

Table 4: Truth Table of $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional Statement:

If p and q are statements, then the compound statement p if and only if q , denoted by $p \leftrightarrow q$ is called a biconditional statement.

$p \leftrightarrow q$ is true when both p and q are true or both p and q are false.

Table 5: Truth Table of $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Ex. Show that $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Sol. By the truth table

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

Here, The 3rd column and the last column have identical truth values. So these are equivalent.

Negation of Compound Statements

Negation of Conjunction: The negation of a conjunction $p \wedge q$ is the disjunction of the negation of p and the negation of q . Equivalently,

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Negation of Disjunction: The negation of a disjunction $p \vee q$ is the conjunction of the negation of p and the negation of q . Equivalently,

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

p	q	$\sim p$	$\sim q$	$\sim p \wedge \sim q$	$p \vee q$	$\sim(p \vee q)$
T	T	F	F	F	T	F
T	F	F	T	F	T	F
F	T	T	F	F	T	F
F	F	T	T	T	F	T

Negation of a Negation: A negation of negation of a statement is the statement itself. Equivalently,

$$\sim(\sim p) \equiv p$$

Negation of Conditional: If p and q are two statements, then

$$\sim(p \rightarrow q) \equiv p \wedge \sim q$$

p	q	$p \rightarrow q$	$\sim(p \rightarrow q)$	$\sim q$	$p \wedge \sim q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

Negation of Biconditional: If p and q are two statements, then

$$\sim(p \leftrightarrow q) \equiv p \leftrightarrow \sim q \equiv \sim p \leftrightarrow q$$

p	q	$p \leftrightarrow q$	$\sim(p \leftrightarrow q)$	$\sim p$	$\sim p \leftrightarrow q$	$\sim q$	$p \leftrightarrow \sim q$
T	T	T	F	F	F	F	F
T	F	F	T	F	T	T	T
F	T	F	T	T	T	F	T
F	F	T	F	T	F	T	F

Exclusive Disjunction: If p and q are two statements, "Either p or q , but not both p and q ". It is denoted by $p \oplus q$.

$p \oplus q$ is not true if both p and q are true or both p & q are false.

Table: Truth Table for $p \oplus q$

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Tautologies: A compound proposition that is always true for all possible truth values of its variables or, in other words contain only T in the last column of its truth table is called a tautology.

For eg: $p \vee \sim p$ ("p or not p") is a tautology as below:

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

Contradictions: A compound proposition that is always false for all possible truth values of its variables or, in other words contain only F in the last column of its truth table is called a contradiction.

For eg: $p \wedge \sim p$ ("p and not p") is a contradiction.

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

Contingency: A proposition that is neither a tautology nor a contradiction is called a contingency.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Tautologically Implication:

A compound proposition $A (p, q, \dots)$ is said to be tautologically imply or simply imply the compound proposition $B (p, q, \dots)$ if and only if $A \rightarrow B$ is a tautology. This is denoted by $A \Rightarrow B$ and read as 'A implies B'.

For eg: $p \wedge q \Rightarrow q$ since $p \wedge q \rightarrow q$ is a tautology which can be seen from the truth table.

p	q	$p \wedge q$	$p \wedge q \rightarrow q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Some important implications which can be proved by truth table.

1, $p \Rightarrow p \vee q$

2, $\sim p \Rightarrow p \rightarrow q$

Some more kinds of Proposition:

1. Converse: The converse of a given conditional is a new conditional formed by interchanging the antecedent and consequent of the given conditional.

If $p \rightarrow q$ then
converse is $q \rightarrow p$.

For eg: If 15 is an odd number, then 3 is a factor of 15.

Converse: If 3 is a factor of 15, then 15 is an odd number.

Truth Table of the Converse

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

2. Inverse: The inverse of a given conditional is a new conditional whose antecedent is the negation of the original antecedent and whose consequent is the negation of the original consequent.

If $p \rightarrow q$ then
inverse is $\sim p \rightarrow \sim q$.

Truth Table of the Inverse

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim p \rightarrow \sim q$
T	T	F	F	T	T
T	F	F	T	F	T
F	T	T	F	T	F
F	F	T	T	T	T

for eg:

Original: If it rains, then I shall not go out.

Inverse: If it does not rain, then I shall go out.

3. Contrapositive: The contrapositive of a given conditional is a new conditional whose antecedent is the negation of the original consequent and whose consequent is the negation of the original antecedent.

Original: $p \rightarrow q$

Contrapositive: $\sim q \rightarrow \sim p$

for eg: Original: If it rains, then I shall not go out.
Contrapositive: If I shall go out, then it does not rain.

Truth Table

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

ARGUMENT $\vdash$:

It is an assertion that the given set of proposition $P_1, P_2, \dots, P_n$ called premises yields another proposition Q called the conclusion and is denoted by

$$P_1, P_2, \dots, P_n \vdash Q.$$

Valid Argument: An argument $P_1, P_2, \dots, P_n \vdash Q$ is said to be valid if Q is true whenever all the premises $P_1, P_2, \dots, P_n$ are true. OR

The argument is valid iff

$$(P_1 \wedge P_2 \wedge \dots \wedge P_n) \rightarrow Q \text{ is a tautology.}$$

Fallacy: It is an argument which is not valid.

NOTE: Always write the premises in terms of $P_1, P_2, \dots, P_n$ and conclusion in Q not small q . because
 $p \rightarrow$ antecedent, $q \rightarrow$ Consequent.

For eg: 1. If I work hard, I shall get a job. I got a job, therefore I worked hard.

Sol. Let p : I work hard.
 q : I get a job.

Premises P_1 : $p \rightarrow q$

P_2 : q

Conclusion Q : p

The symbolic form of the argument

$$p \rightarrow q, q \vdash p$$

Now to check its validity we will write its argument as

$p \rightarrow q \wedge q \rightarrow p$ should be tautology.

p	q	$p \rightarrow q$	$p \rightarrow q \wedge q$	$(p \rightarrow q \wedge q) \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	[F]
F	F	T	F	T

Since it is not a tautology, hence it is not a valid argument.

Ex.2. Find if the following argument is valid or not.

If it rains, he will be sick

It did not rain. Therefore, ~~it did not rain.~~
he was not sick.

Sol. Let p : It rains,
 q : He is sick.

Premises: P_1 : $p \rightarrow q$

P_2 : $\sim p$

Conclusion: Q : $\sim q$

In symbolic form the argument is

$p \rightarrow q, \sim p \vdash \sim q$

To check the validity,

$(p \rightarrow q) \wedge \sim p \rightarrow \sim q$ should be tautology.

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$p \rightarrow q \wedge \sim p$	$(p \rightarrow q) \wedge \sim p \rightarrow \sim q$
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	T	[F]
F	F	T	T	T	T	T

Since it is not a tautology, hence it is not a valid argument.

Ex.3. If Ram works hard, he will get a job.

Ram works hard therefore he will get a job.

Sol. Let p : Ram works hard

q : He will get a job.

Premises $P_1: p \rightarrow q$

$P_2: p$

Conclusion $Q: q$

In symbolic form the argument is

$$p \rightarrow q, p \vdash q$$

Now will check its validity.

$$(p \rightarrow q) \wedge p \rightarrow q$$

p	q	$p \rightarrow q$	$p \rightarrow q \wedge p$	$(p \rightarrow q) \wedge p \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

It shows tautology so it is a valid argument.

Ex.4. Check validity of:

Either Ram will cook or Hari will practice Karate. If Hari practices Karate, then Ram studies, Ram does not study. Therefore Hari will practice Karate.

Sol. Let p : Ram will cook

q : Hari will practice Karate

r : Ram studies

The symbolic form of the argument is

$$(p \vee q), q \rightarrow r, \neg r \vdash q$$

Now The truth table is

p	q	r	$\sim r$	$p \vee q$	$q \rightarrow r$	$\overset{A}{(p \vee q) \wedge (q \rightarrow r) \wedge \sim r}$	$A \rightarrow q$
T	T	T	F	T	T	F	T
T	T	F	T	T	F	F	T
T	F	T	F	T	T	F	T
T	F	F	T	T	T	T	F
F	T	T	F	T	T	F	T
F	T	F	T	T	F	F	T
F	F	T	F	F	T	F	T
F	F	F	T	F	T	F	T

It does not shows the tautology. Hence the argument is not valid.

In case the no. of propositions become more and the truth table method cannot be used we are bound to use some other method and that some other method is provided to us. So these are the inference rules.

Rules of Inference

Inference Rule	Tautology	Name
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens (mode that affirms)
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens (mode that denies)
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge (\neg p)) \rightarrow q$	disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$(p \wedge q) \rightarrow (p \wedge q)$	conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	resolution

Modus Ponens:

1. P Premise
2. $P \rightarrow Q$ Premise
3. Q Modus ponens (1,2)

It is not necessary that these premises be simple atomic proposition, they can be compound proposition also. As shown in this example:

1. $(A \vee \sim B) \rightarrow \sim C$ Premise
2. $\sim D$ Premise
3. $A \vee \sim B$ Premise
4. $\sim C$ Modus ponens (1,3)

Ex.

1. P Premise
2. $(P \wedge Q) \rightarrow R$ Premise
3. R INVALID modus ponens!

Modus Tollens:

1. $\sim Q$ Premise
2. $P \rightarrow Q$ Premise
3. $\sim P$ Modus tollens (1,2)

Ex.

1. $\sim(A \rightarrow B)$ Premise
2. $C \rightarrow (A \rightarrow B)$ Premise
3. $\sim C$ Modus tollens (1,2)

Hypothetical Syllogism:

1. $P \rightarrow (Q \vee R)$ Premise
2. $(Q \vee R) \rightarrow \sim S$ Premise
3. $P \rightarrow \sim S$ Rule of Syllogism (1,2)

Disjunctive Syllogism:

1. $\sim P$ Premise
2. $P \vee Q$ Premise
3. Q Disjunctive Syllogism (1,2)

1. $\sim (A \vee B)$

Premise

2. $(A \vee B) \vee D$

Premise

3. D

Disjunctive syllogism (1,2)

Which rule of inference is used in each argument below?

• Alice is a Math major. Therefore, Alice is either a Math major or a CSI major. $p \vdash p \vee q$ Addition

• Jerry is a Math major and a CSI major. Therefore, Jerry is a Math major. $p \wedge q \vdash p$ Simplification

• If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed. Modus Ponens

$p \rightarrow q, p \vdash q$

• If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.

$p \rightarrow q, \sim q \vdash \sim p$

Modus Tollens

• If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.

$p \rightarrow q, q \rightarrow r \vdash p \rightarrow r$

Hypothetical Syllogism

• I go swimming or eat an ice cream. I did not go swimming. Therefore, I eat an ice cream.

$p \vee q, \sim p \vdash q$

Disjunctive Syllogism

Ex. Given $x, x \rightarrow y, y \rightarrow z, \sim z \vee w$, prove w .

Sol

1. $x \rightarrow y$

Premise

2. $y \rightarrow z$

Premise

3. $x \rightarrow z$

Hypothetical Syllogism (1,2)

4. x

Premise

5. z

Modus Ponens (3,4)

6. $\sim Z \vee W$

Premise

7. W

Disjunctive Syllogism (5,6)

Example: Premises: $\begin{cases} \sim A \rightarrow (C \wedge D) \\ A \rightarrow B \\ \sim B \end{cases}$ Prove: C .

Sol.

1. $A \rightarrow B$	Premise
2. $\sim B$	Premise
3. $\sim A$	Modus tollens (1,2)
4. $\sim A \rightarrow (C \wedge D)$	Premise
5. $C \wedge D$	Modus ponens (3,4)
6. C	Simplification (5)

Valid Arguments.

Example: Given the premises: $p, p \rightarrow q, s \vee r, r \rightarrow \sim q$
Arrive at the conclusion: $s \vee t$

Sol.

1. p	Premise
2. $p \rightarrow q$	Premise
3. q	Modus Ponens (1,2)
4. $r \rightarrow \sim q$	Premise
5. $q \rightarrow \sim r$	Contrapositive statement of 4.
6. $\sim r$	Modus Ponens (3,5)
7. $s \vee r$	Premise
8. s	Disjunctive Syllogism (6,7)
9. $s \vee t$	Addition (8).

Example: Premises: $\begin{cases} P \wedge Q \\ P \rightarrow \sim (Q \wedge R) \\ S \rightarrow R \end{cases}$

Prove: $\sim S$.

- | | |
|-------------------------------------|------------------------------|
| 1. $P \wedge Q$ | Premise |
| 2. P | Simplification (1) |
| 3. Q | Simplification (1) |
| 4. $P \rightarrow \sim(Q \wedge R)$ | Premise |
| 5. $\sim(Q \wedge R)$ | Modus Ponens (2, 4) |
| 6. $\sim Q \vee \sim R$ | De Morgan (5) |
| 7. $\sim R$ | Disjunctive Syllogism (3, 6) |
| 8. $S \rightarrow R$ | Premise |
| 9. $\sim S$ | Modus Tollens (7, 8) |

Example. Premises: $\begin{cases} \sim(A \vee B) \rightarrow C \\ \sim A \\ \sim C \end{cases}$

Prove: B.

- | | |
|-----------------------------------|------------------------------|
| 1. $\sim(A \vee B) \rightarrow C$ | Premise |
| 2. $\sim C$ | Premise |
| 3. $A \vee B$ | Modus tollens (1, 2) |
| 4. $\sim A$ | Premise |
| 5. B | Disjunctive syllogism (3, 4) |

Example: From the single proposition
 $p \wedge (p \rightarrow q)$

show that q is a conclusion.

Sol.

- | <u>Step</u> | <u>Reason</u> |
|---------------------------------|---------------------|
| 1. $p \wedge (p \rightarrow q)$ | Premise |
| 2. p | Simplification (1) |
| 3. $p \rightarrow q$ | Simplification (1) |
| 4. q | Modus Ponens (2, 3) |

Example: Suppose $P \rightarrow Q$; $\sim P \rightarrow R$; $Q \rightarrow S$.
Prove that $\sim R \rightarrow S$.

- Sol.
- | | | |
|----|------------------------------------|------------------------------|
| 1. | $P \rightarrow Q$ | Premise |
| 2. | $\sim P \vee Q$ | Logically equivalent (1) |
| 3. | $\sim P \rightarrow R$ | Premise |
| 4. | $P \vee R$ | Logically equivalent (3) |
| 5. | $R \vee Q$ | Resolution (2,4) |
| 6. | $\sim R \rightarrow Q$ | Logically equivalent (5) |
| 7. | $Q \rightarrow S$ | Premise |
| 8. | $\sim R \rightarrow \underline{S}$ | Hypothetical syllogism (6,7) |

Example: Show that the following is a valid argument:

If it rains, I don't go to work. If I don't go to work, I get fired. I'm not getting fired today. Therefore, it is not raining.

Sol.

p = It rains.
 q = I go to work
 r = I get fired.

- | | | |
|----|---------------------------------|------------------------------|
| 1. | $p \rightarrow \sim q$ | Premise |
| 2. | $\sim q \rightarrow r$ | Premise |
| 3. | $\sim r$ | Premise |
| 4. | $p \rightarrow r$ | Hypothetical syllogism (1,2) |
| 5. | $\therefore \underline{\sim p}$ | Modus tollens (3,4) |

So it is valid argument.

Example:

- With these hypotheses:

"It is not sunny this afternoon and it is colder than yesterday."

"We will go swimming only if it is sunny."

"If we do not go swimming, then we will take a ~~can~~ boat trip."

"If we take a boat trip, then we will be home by sunset."

- Using the inference rules, construct a valid argument for the conclusion:

"We will be home by sunset."

Sol. Consider the following propositions:

p : It's sunny this afternoon.

q : It's colder than yesterday.

r : We will go swimming

s : We will take a boat trip.

t : We will be home by sunset.

Premises:

a. $\neg p \wedge q$

b. $r \rightarrow p$

c. $\neg r \rightarrow s$

d. $s \rightarrow t$

Conclusion: t .

- | | | |
|----|-------------------|---------------------|
| 1. | $\neg p \wedge q$ | Premise |
| 2. | $\neg p$ | Simplification (1) |
| 3. | $r \rightarrow p$ | Premise |
| 4. | $\neg r$ | Modus tollens (2,3) |

- | | | |
|----|------------------------|---------------------|
| 5. | $\sim r \rightarrow s$ | Premise |
| 6. | s | Modus Ponens (4, 5) |
| 7. | $s \rightarrow t$ | Premise |
| 8. | t | Modus Ponens (6, 7) |

This is a valid argument showing that from the premises (a), (b), (c) and (d), we can prove the conclusion t .

Example: If it is Saturday today, then we play soccer or basketball.

If the soccer field is occupied, we don't play soccer.

It is Saturday today, and the soccer field is occupied.

Prove: "We play basketball or volleyball."

Sol p : It is Saturday today.

q : We play soccer.

r : We play basketball.

s : The soccer field is occupied.

t : We play volleyball.

Premises: $p \rightarrow (q \vee r)$, $s \rightarrow \sim q$, $p \wedge s$

Need to prove: $r \vee t$.

- | | | |
|----|----------------------------|------------------------------|
| 1. | $p \rightarrow (q \vee r)$ | Premise |
| 2. | $p \wedge s$ | Premise |
| 3. | p | Simplification (2) |
| 4. | $q \vee r$ | Modus Ponens (1, 3) |
| 5. | $s \rightarrow \sim q$ | Premise |
| 6. | s | Simplification (2) |
| 7. | $\sim q$ | Modus Ponens (5, 6) |
| 8. | r | Disjunctive Syllogism (4, 7) |
| 9. | $r \vee t$ | Addition (8) |

So Valid Argument.

Example: Show that:

"If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on. If the sailing race is held, then the trophy will be awarded. The trophy was not awarded." implies "It rained."

Sol. p : It rains

q : It is foggy

r : sailing race will be held

s : Lifesaving Demo will go on.

t : The trophy will be awarded.

- | | | |
|----|---|---------------------|
| 1. | $(\sim p \vee \sim q) \rightarrow (r \wedge s)$ | Premise |
| 2. | $r \rightarrow t$ | Premise |
| 3. | $\sim t$ | Premise |
| 4. | $\sim r$ | Modus Tollens (2,3) |
| 5. | $\sim r \vee \sim s$ | Addition (4) |
| 6. | $\sim (r \wedge s)$ | De-Morgan's law (5) |
| 7. | $\sim (\sim p \vee \sim q)$ | Modus tollens (1,6) |
| 8. | $p \wedge q$ | De-Morgan's law (7) |
| 9. | p | Simplification (8) |

Mathematical Induction

Suppose we have a statement that says something about all natural numbers n .

Let's call this statement P .

- For example,

P : for every natural number n , the sum of the first n odd numbers is n^2 .

Since this statement is about all natural numbers, it contains infinitely many statements.

- We will call them $P(1)$, $P(2)$,

$P(1)$: the sum of the first 1 odd number is 1^2 .

$P(2)$: the sum of the first 2 odd numbers is 2^2 .

$P(3)$: the sum of the first 3 odd numbers is 3^2 .

⋮

⋮

How can we prove all of these statements at once?

- Mathematical induction is a clever way of doing so.

The crux of the idea is:

- Suppose we can prove that, whenever one of these statements is true, then the one following it in the list is also true.

- That is,

for every k , if $P(k)$ is true, then $P(k+1)$ is true

- This is called the induction step since it leads us from the truth of one statement to the next.

Now, suppose that we can also prove that $P(1)$ is true.

- The induction step now leads us through the following chain of statements.

- $P(1)$ is true, so $P(2)$ is true.

$P(2)$ is true, so $P(3)$ is true.

$P(3)$ is true, so $P(4)$ is true.

$\vdots$

So, we see that, if both the induction step and $P(1)$ are proved, then statement P is proved for all n .

Principle of Mathematical Induction

To apply this principle, there are two steps:

1. Prove that $P(1)$ is true.
2. Assume that $P(k)$ is true and use this assumption to prove that $P(k+1)$ is true.

Example: Prove that, for all natural numbers n ,

$$1 + 3 + 5 + \dots + (2n-1) = n^2$$

Sol. Let $P(n)$ denote the statement.

Step 1. We need to show that $P(1)$ is true.

- However, $P(1)$ is simply the statement that $1 = 1^2$.
- This is, of course, true.

Step 2. We assume that $P(k)$ is true.

- Thus, our induction hypothesis is:

$$1 + 3 + 5 + \dots + (2k-1) = k^2$$

We want to use this to show that $P(k+1)$ is true.

$$1 + 3 + 5 + \dots + (2k-1) + [2(k+1)-1] = (k+1)^2$$

We start with the left side and use the induction hypothesis to obtain the right side of the equation:

$$1 + 3 + 5 + \dots + (2k-1) + [2(k+1)-1]$$

$$= k^2 + (2(k+1)-1)$$

$$= k^2 + (2k+2-1)$$

$$= k^2 + 2k + 1$$

Induction Hypothesis

$$= (k+1)^2$$

- Thus, $P(k+1)$ follows from $P(k)$.
- This completes the induction step.

Having proved steps 1 and 2, we conclude by the Principle of Mathematical Induction that $P(n)$ is true for all natural numbers n .

Example 2: Prove that, for every natural number n ,

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Proof Let $P(n)$ be the statement.

Step 1. We need to show that $P(1)$ is true.

- However, $P(1)$ says that:

$$1 = \frac{1(1+1)}{2}$$

- This statement is clearly true.

Step 2. Assume that $P(k)$ is true.

- Thus, our induction hypothesis is:

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

We want to use this to show that $P(k+1)$ is true.

$$\bullet \quad 1 + 2 + 3 + \dots + k + (k+1) = \frac{(k+1)[(k+1)+1]}{2}$$

$$(1 + 2 + 3 + \dots + k) + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= (k+1) \left(\frac{k}{2} + 1 \right) = (k+1) \left(\frac{k+2}{2} \right)$$

$$= \frac{(k+1)[(k+1)+1]}{2}$$

- Thus, $P(k+1)$ follows from $P(k)$.

We conclude that $P(n)$ is true for all natural numbers n .

Sums of Powers

Formulas for the sums of powers of the first n natural numbers.

$$\sum_{k=1}^n 1 = n$$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$$

Formula 2 is proved in Example 2.

The others are also proved using mathematical induction.

Example 3: Prove that $4n < 2^n$ for all $n \geq 5$

Sol. Let $P(n)$ denote the statement $4n < 2^n$.

Step 1. $P(5)$ is the statement that

$$4 \cdot 5 < 2^5, \quad \text{or} \quad 20 < 32.$$

This is true.

Step 2. Assume that $P(k)$ is true.

• Thus, our induction hypothesis is:

$$4k < 2^k$$

• We want to use this to show $P(k+1)$ is true.

$$4(k+1) < 2^{k+1}$$

$$\text{L.H.S.} = 4(k+1) = 4k + 4$$

$$< 2^k + 4$$

$$< 2^k + 4k \quad (\text{Since } 4 < 4k)$$

$$< 2^k + 2^k$$

$$< 2 \cdot 2^k < 2^{k+1}$$

Thus, $P(k+1)$ follows from $P(k)$.

So $P(n)$ is true for all natural numbers $n \geq 5$.

Example: 4. Prove that, for every natural number n ,
 $7^{2n} + 2^{3n-3} \cdot 3^{n-1}$ is divisible by 25.

Sol. Let $P(n)$ be the statement.

We have to show that $P(n)$ is true for all natural no. n .

We have to show that 25 divides $P(n)$ i.e.,

$$P(n) = 25m \quad \forall m \in \mathbb{Z}$$

$$P(n) = 49^n + 8^{n-1} \cdot 3^{n-1}$$

$$= 49^n + 24^{n-1}$$

Step 1. We need to show that $P(1)$ is true.

$P(1)$ says that:

$$49 + 1 = 50 = 25 \times 2$$

So 25 divides $P(1)$

• This statement is clearly true.

Step 2. Assume that $P(k)$ is true.

Thus, our induction hypothesis is:

$$49^k + 24^{k-1} = 25m$$

We want to use this to show that $P(k+1)$ is true.

That is,

$49^{k+1} + 24^k$ is divisible by 25.

So, we use the induction hypothesis to obtain this:

$$49^{k+1} + 24^k = 49^k \cdot 49 + 24^k$$

$$= 49(25m - 24^{k-1}) + 24^k$$

$$= 49 \times 25m - 49 \times 24^{k-1} + 24^k$$

$$= 49 \times 25m - 24^{k-1}(49 - 24)$$

$$= 49 \times 25m - 24^{k-1} \times 25$$

$$= 25(49m - 24^{k-1})$$

$$= 25 \cdot q \quad \text{where } q = 49m - 24^{k-1}$$

Thus, $P(k+1)$ follows from $P(k)$.

We conclude that $P(n)$ is true for all natural nos n .

Example 5. Prove by mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n .

Sol. Let $P(n)$ be the given statement.

1. Inductive base: For $n=1$,

$$6^{1+2} + 7^{2+1} = 6^3 + 7^3 = 216 + 343 \\ = 559 = 43 \cdot 13$$

which is divisible by 43. So $P(1)$ is true.

2. Inductive hypothesis:

Assume that $P(k)$ is true i.e.,

$$6^{k+2} + 7^{2k+1} = 43 \cdot m \quad \text{for some integer } m.$$

3. Inductive step:

We wish to show the truth of $P(k+1)$

$$6^{(k+1)+2} + 7^{2(k+1)+1} = 6^{k+3} + 7^{2k+3} \text{ is divisible by 43}$$

$$\begin{aligned} \text{Now, } 6^{k+3} + 7^{2k+3} &= 6 \cdot 6^{k+2} + 7^{2k+3} \\ &= 6 \cdot (43m - 7^{2k+1}) + 7^{2k+1} \cdot 7^2 \\ &= 6 \times 43m - 6 \times 7^{2k+1} + 49 \times 7^{2k+1} \\ &= 6 \times 43m + 7^{2k+1} (49 - 6) \\ &= 6 \times 43m + 7^{2k+1} \times 43 \\ &= 43 (6m + 7^{2k+1}) \end{aligned}$$

which is divisible by 43.

Hence $P(k+1)$ is true whenever $P(k)$ is true. So $P(n)$ is true for all positive integer of n .

Q1. Show that $n^2 > 2n+1$ for $n \geq 3$ by mathematical induction.

Q2. Use Induction to show that

$$n! \geq 2^{n-1} \text{ for } n \geq 1$$

Example 6. Show that

$$\frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

Sol. Let the given statement be $P(n)$.

1. Inductive base: For $n=1$.

$$\frac{1}{1.2} = \frac{1}{1+1} = \frac{1}{2} \quad \text{Hence } P(1) \text{ is true.}$$

2. Inductive hypothesis:

Assume that $P(k)$ is true.

$$\frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{k.(k+1)} = \frac{k}{k+1}$$

3. Inductive step: We wish to show that the statement is true for $n=k+1$ i.e.,

$$\frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{(k+1).(k+2)} = \frac{k+1}{k+2}$$

$$\begin{aligned} \text{Now, } & \frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k^2 + 2k + 1}{(k+1)(k+2)} \\ &= \frac{(k+1)}{k+2} \end{aligned}$$

Thus $P(k+1)$ is true.

So $P(n)$ is true for all positive integer n .

Prove by the principle of mathematical induction
 $n! > 2^n$ for all $n \geq 4$

Proof. Let $P(n) : n! > 2^n$

$$4! = 1 \cdot 2 \cdot 3 \cdot 4$$

$$= 24 > 2^4 \text{ is true}$$

and thus $P(4)$ is true.

Now suppose that the statement is true for $n=k$, $k \geq 4$
 $k! > 2^k$.

We have to prove that the statement is true for $n=k+1$,
we must prove that

$$(k+1)! > 2^{k+1}$$

Now

$$\begin{aligned} (k+1)! &= (k+1) k! \\ &> (k+1) 2^k \end{aligned}$$

Since for $k \geq 4$, $k+1 > 2$

Hence,

$$(k+1) 2^k > 2 \cdot 2^k = 2^{k+1}$$

Hence, $(k+1)! > 2^{k+1}$ as desired, $P(k+1)$ is true.

Thus, by the principle of mathematical induction,
 $n! > 2^n$ for all integers $n \geq 4$.